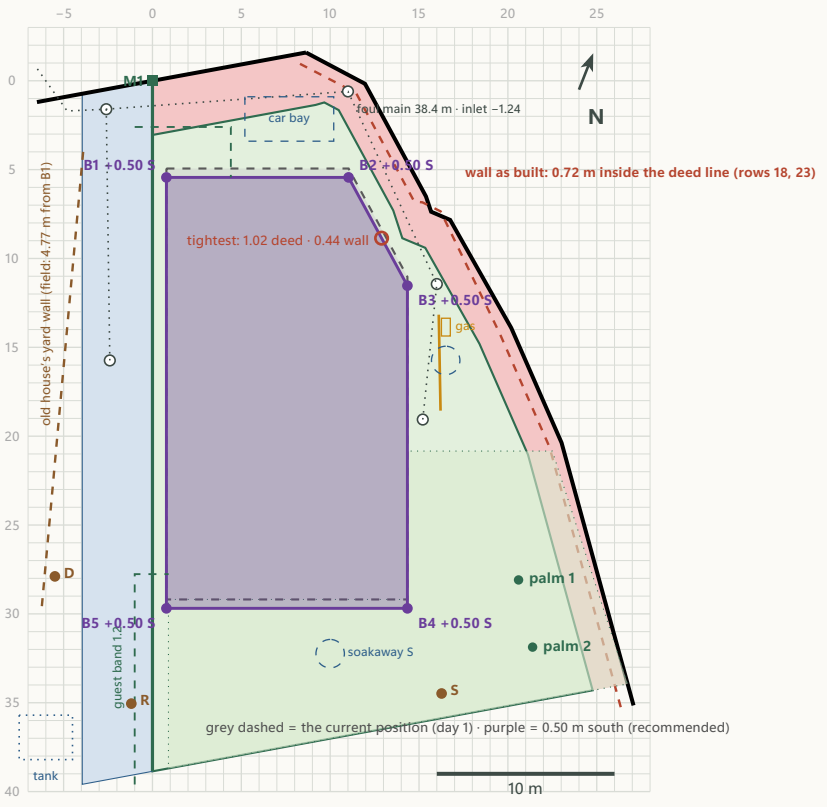


Shift SOUTH 0.40 / 0.50 / 0.60 m · study

Start: the current position (day 1: B1 5.00 m from M1, the west face 0.79 m east of the M line). Calibration: B2 2.40 m and the cut wall's tightest point 2.20–2.30 m to the wall's inner face (rows 23 and 18) → the wall as built 0.72 m inside the deed line (the lower reading; the 2.30 reading gives 0.62); B1 4.77 m to the yard wall. Every wall sampled at 5 cm. Target: ≥ 0.40 m from the wall as built.

South by (m)	**current**	**0.40**	**0.50** (rec.)	**0.60**
Tightest margin on any wall, from the deed line (the cut wall at the lane's notch; ≥ 0.50 wanted)	0.79	0.98	1.02	1.07
Tightest margin from the wall as built: the 2.20 reading / the 2.30 reading (target ≥ 0.40)	0.20 / 0.30	0.39 / 0.49	0.44 / 0.54	0.49 / 0.59
Where: on the cut wall at (x, y) · its distance to the deed line there	(13.1, 8.8) · 2.92	(12.9, 8.8) · 3.11	(12.9, 8.9) · 3.16	(12.9, 8.9) · 3.21
B2's margin: deed / from the wall · B3 to the deed line / to the wall	1.08 / 0.36 · 3.67 / 2.95	1.27 / 0.55 · 3.87 / 3.15	1.32 / 0.60 · 3.91 / 3.19	1.37 / 0.65 · 3.96 / 3.24
M1 → B1 (the owner's 5.00 rule) · north strip: boundary wall to the house at B1 · to the north setback line	**5.00** · 5.08 · 2.00	**5.40** · 5.48 · 2.39	**5.49** · 5.58 · 2.49	**5.59** · 5.68 · 2.59
Car bay (kept where it is): gap to the north wall · the swept path's last move clear of the wall · the reverse move past B1	1.54 · 1.89 · 0.46	1.94 · 2.29 · 0.74	2.04 · 2.39 · 0.81	2.14 · 2.49 · 0.88
Family entrance path from the driveway (m) · guest path from the driveway's south end (m)	7.75 · 14.25	8.15 · 13.85	8.25 · 13.75	8.35 · 13.65
West wall to the driveway's west edge (unchanged) · to the yard wall at B1 / y 15 / y 25 / B5	4.74 · 4.77 / 5.68 / 6.59 / 6.97	4.74 · 4.81 / 5.68 / 6.59 / 7.01	4.74 · 4.82 / 5.68 / 6.59 / 7.02	4.74 · 4.82 / 5.68 / 6.59 / 7.03
Driveway clear width · the cornice's edge east of the M line · guest band 1.2 + strip 0.79	3.95 · 0.49 · 1.99	3.95 · 0.49 · 1.99	3.95 · 0.49 · 1.99	3.95 · 0.49 · 1.99
Palm 1 / palm 2 to the nearest wall	5.99 / 7.27	5.99 / 7.14	5.99 / 7.11	5.99 / 7.08
South family garden (m^2) · lost against the current	**260** · 0	**255** · −5.38	**253** · −6.72	**252** · −8.07
Outdoor kitchen, screen, gas, grill move with the house (spacings unchanged) · gas to the wall as built (≥ 1.5)	1.75	1.95	2.00	2.05
Existing tank to B5 (≥ 5) · B5 to the garden's low wall	8.39 · 0.19	8.08 · 0.19	8.00 · 0.19	7.93 · 0.19
Sewer: the east main's length (m) · the inlet at the septic on the assumed falls (was −1.23) · the west run to its chamber	37.9 · −1.23 · 3.31	38.3 · −1.24 · 3.33	38.4 · −1.24 · 3.35	38.5 · −1.24 · 3.36
Soakaway E to the east wall · soakaway S to the south wall	2.15 · 3.05	2.15 · 2.65	2.15 · 2.55	2.15 · 2.45
Cut wall: how much it could later move outwards keeping 0.40 from the wall (2.20 / 2.30 reading)	0.00 / 0.00	0.00 / 0.09	0.04 / 0.14	0.09 / 0.19
… and the floor area gained per floor (m^2, 2.20 / 2.30 reading)	0.00 / 0.00	0.00 / 0.62	0.27 / 0.94	0.59 / 1.25
Conflicts	none	none	none	none

Moving south adds 0.05 m of margin at the cut wall per 0.10 m (the cut runs at 61° to the M line); nothing moves in x, so the driveway, the west strip and the route's width are untouched. The cut-wall rows: a parallel outward move of d shortens the cut wall and adds ≈ 6.9 d m^2 per floor: the air tower ≈ 1.14 d, the maid bath 2.05 d, the maid room 3.64 d (ground floor); the tower base 1.14 d and bedroom 1 5.69 d (first floor). Bedroom 1's hotel en-suite B4 needs 15.74 m^2 clear (now 13.98) and a rectangular room: it never fits with these moves.

Gains and costs · what each option gives and takes, against the current position

0.40 m south

B1 5.40 m from M1 · north strip 5.48 m deep at B1

I GAIN: a margin of **0.39 m from the wall as built** at the cut wall (0.49 if the 2.30 reading holds; today 0.20–0.30), 0.98 m from the deed line (today 0.79) — the target is met only at the 2.30 reading B2 0.55 m from the wall (today 0.36). Room to shorten the cut wall later: **0.00–0.09 m** (0.00–0.62 m² per floor: the maid room, the maid bath and the air tower below, bedroom 1 and the tower base above; the en-suite still does not fit). The north strip 5.48 m deep at B1 (today 5.08): the car bay keeps 1.94 m to the wall (today 1.54) and the reverse move past B1 clears 0.74 m (today 0.46): the 3-point turn stays, with 0.40 m more room; the bay could itself move 0.40 m south if the owner wishes. The route beside the driveway: unchanged in width (4.81 m at B1). **I LOSE: 5 m² of the south garden** (260 → 255 m², 2.1 %), the strip between the house and the palms: palm 2 7.27 → 7.14 m (palm 1 unchanged, 5.99); the family walk from the gate 7.75 → 8.15 m (+0.40); the guest walk 14.25 → 13.85 m (−0.40); the south soakaway 3.05 → 2.65 m from the south wall; the tank 8.39 → 8.08 m from B5 (still ≥ 5); the foul main 37.9 → 38.3 m and the inlet −1.23 → −1.24 (nothing in practice); the owner's rule "B1 = 5.00 m" becomes 5.40 all five blocks re-set. **The extra 0.10 m (from the current position):** +0.19 m of margin for 5 m² of garden. **Verdict:** **not enough: 0.39 m from the wall at the lower reading (0.49 at 2.30) — the target is missed unless the 2.30 reading holds**.

0.50 m south

B1 5.49 m from M1 · north strip 5.58 m deep at B1

I GAIN: a margin of **0.44 m from the wall as built** at the cut wall (0.54 if the 2.30 reading holds; today 0.20–0.30), 1.02 m from the deed line (today 0.79) — **the 0.40 m target is met at both readings, without waiting for the engineering office** B2 0.60 m from the wall (today 0.36). Room to shorten the cut wall later: **0.04–0.14 m** (0.27–0.94 m² per floor: the maid room, the maid bath and the air tower below, bedroom 1 and the tower base above; the en-suite still does not fit). The north strip 5.58 m deep at B1 (today 5.08): the car bay keeps 2.04 m to the wall (today 1.54) and the reverse move past B1 clears 0.81 m (today 0.46): the 3-point turn stays, with 0.50 m more room; the bay could itself move 0.50 m south if the owner wishes. The route beside the driveway: unchanged in width (4.82 m at B1). **I LOSE: 7 m² of the south garden** (260 → 253 m², 2.6 %), the strip between the house and the palms: palm 2 7.27 → 7.11 m (palm 1 unchanged, 5.99); the family walk from the gate 7.75 → 8.25 m (+0.50); the guest walk 14.25 → 13.75 m (−0.50); the south soakaway 3.05 → 2.55 m from the south wall; the tank 8.39 → 8.00 m from B5 (still ≥ 5); the foul main 37.9 → 38.4 m and the inlet −1.23 → −1.24 (nothing in practice); the owner's rule "B1 = 5.00 m" becomes 5.49 all five blocks re-set. **The extra 0.10 m (from 0.40):** +0.05 m of margin for 1 m² of garden: **this is the step that reaches the target**. **Verdict:** **RECOMMENDED: the target met at both readings (0.44 / 0.54) within the ground the owner checked, for 7 m² of garden**.

0.60 m south

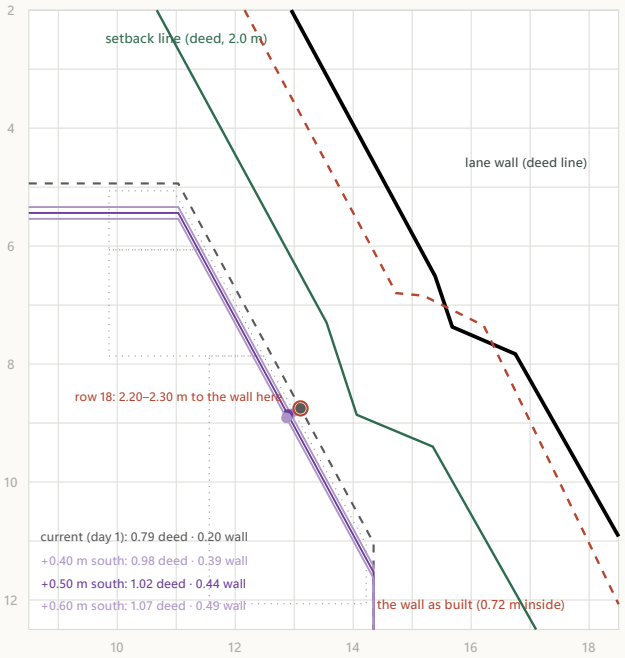
B1 5.59 m from M1 · north strip 5.68 m deep at B1

I GAIN: a margin of **0.49 m from the wall as built** at the cut wall (0.59 if the 2.30 reading holds; today 0.20–0.30), 1.07 m from the deed line (today 0.79) — **the 0.40 m target is met at both readings, without waiting for the engineering office** B2 0.65 m from the wall (today 0.36). Room to shorten the cut wall later: **0.09–0.19 m** (0.59–1.25 m² per floor: the maid room, the maid bath and the air tower below, bedroom 1 and the tower base above; the en-suite still does not fit). The north strip 5.68 m deep at B1 (today 5.08): the car bay keeps 2.14 m to the wall (today 1.54) and the reverse move past B1 clears 0.88 m (today 0.46): the 3-point turn stays, with 0.60 m more room; the bay could itself move 0.60 m south if the owner wishes. The route beside the driveway: unchanged in width (4.82 m at B1). **I LOSE: 8 m² of the south garden** (260 → 252 m², 3.1 %), the strip between the house and the palms: palm 2 7.27 → 7.08 m (palm 1 unchanged, 5.99); the family walk from the gate 7.75 → 8.35 m (+0.60); the guest walk 14.25 → 13.65 m (−0.60); the south soakaway 3.05 → 2.45 m from the south wall; the tank 8.39 → 7.93 m from B5 (still ≥ 5); the foul main 37.9 → 38.5 m and the inlet −1.23 → −1.24 (nothing in practice); the owner's rule "B1 = 5.00 m" becomes 5.59 all five blocks re-set. **The extra 0.10 m (from 0.50):** +0.05 m of margin for 1 m² of garden: only the cut-wall option grows (0.04 → 0.09 m), the margin is already met. **Verdict:** **only if the ground really allows 0.60: +0.05 m more margin and a 0.09–0.19 m cut-wall option, for another 1 m² of garden**.

Recommendation: move 0.50 m south

The current position keeps only 0.20–0.30 m from the wall as built at the cut wall (row 18), below the owner's 0.40 m target; 0.50 m south is the smallest step that meets it at both readings (0.44 / 0.54) and it lies within the ground the owner checked. It costs 7 m² of garden (2.6 %) and nothing else of substance: the driveway, the west strip, the route, the guest band, the car bay, the yard, the gas cabinet, the tanks, the sewer and the soakaways all keep their rules. 0.40 m misses the target at the lower reading (0.39); 0.60 m adds 0.05 m and a 0.09–0.19 m cut-wall option for 1 m² more garden and ground the owner is not sure of. **So 0.50 m south is clearly better than staying; more is not.** The cut wall stays 6.93 m for now (decision, row 18); the official survey has the last word.

الخلاصة بالعربية · إزاحة البيت جنوبًا: الموضع الحالي يترك عند جدار القطع 0.20–0.30 م فقط من جدار الحد كما هو مبني (قياس الموقع، الصف ١٨)، أي أقل من الهدف 0.40 م. الإزاحة 0.50 م جنوبًا هي أصغر خطوة تحقق الهدف بالقراءتين (0.44 / 0.54 م) وضمن الأرض التي فحصها المالك؛ ثمنها 7 م² من الحديقة الجنوبية (2.6 ٪) وزيادة 0.50 م في مشي العائلة من البوابة؛ لا يتغير الممر ولا الشريط الغربي ولا موقف السيارة ولا الخزانات ولا الصرف. الخيار ٠٫٤٠ م لا يبلغ الهدف بالقراءة الأدنى (0.39 م)، والخيار ٠٫٦٠ م يزيد الهامش 0.05 م فقط ويفتح تقصير جدار القطع 0.09–0.19 م مقابل 1 م² أخرى من الحديقة وأرض غير مؤكدة. **التوصية: 0.50 م جنوبًا.** جدار القطع يبقى ٦٫٩٣ م الآن، والمساحة الرسمية هي الكلمة الأخيرة.



The cut wall enlarged. Grey dashed: the current position; purple: the options (the recommended one bold). The red circle: the model's nearest point to the lane line, where row 18 read 2.20–2.30 m to the wall (about 1 m before B3 by the owner; 2.1 m by the sheet: tape it from M1 16.25 / D 26.53 next time). Each dot = the tightest point of that option. Dotted: the rooms along the cut (the air tower, the maid bath, the maid room).

Set-out table of the recommended option · 0.50 m south

every block moves straight south along the M line

Corner	x east	y along M	Tape 1 (M1)	Tape 2 (D)	Tape 3 (R)	Tape 4 (S)	The block moves from the day-1 position
B1	0.79	5.44	M1 → **5.49**	D → **23.31**	R → **29.68**	S → **32.91**	0.50 m south, x unchanged
B2	11.04	5.44	M1 → **12.30**	D → **27.88**	R → **32.04**	S → **29.51**	0.50 m south, x unchanged
B3	14.35	11.53	M1 → **18.41**	D → **25.71**	R → **28.19**	S → **23.03**	0.50 m south, x unchanged
B4	14.35	29.69	M1 → **32.97**	D → **19.92**	R → **16.44**	S → **5.16**	0.50 m south, x unchanged
B5	0.79	29.69	M1 → **29.70**	D → **6.53**	R → **5.71**	S → **16.21**	0.50 m south, x unchanged

Side / diagonal	m (unchanged)
B1–B2	**10.25**
B2–B3	**6.93**
B3–B4	**18.16**
B4–B5	**13.56**
B5–B1	**24.25**
B1–B4	**27.78**
B2–B5	**26.33**
B1–B3 (the cut's check)	**14.87**
Perimeter	**73.15**

On site: string M1 → M4 as on day 1. Every block moves **0.50 m straight south** along the string direction, x unchanged: from each present block measure 0.50 m parallel to the string (square off with 3-4-5 if the block is away from the string) and drive the new peg; then check the sides and the diagonals (unchanged: a move of all five by the same vector keeps every one) and the tapes above from M1, D, R and S. B1 is then 5.49 m from M1 (the owner's 5.00 rule becomes 5.49); M1 → D, M4 and the palms are unchanged. **Then:** `design_gf.typ` gets the shift (0.91, 5.56) and the set-out sheet and the services plan are re-issued; the garden, the yard, the hedge and the chambers attached to the house move with it, the gate, the driveway, the car bay, the tanks and the septic stay (not done here: the owner decides first). **Cut wall after this move:** 0.44–0.54 m from the wall as built; it stays 6.93 m (decision, row 18); a later outward move of 0.04–0.14 m would keep the 0.40 m and gain 0.27–0.94 m² per floor, after the official survey.

Shift south study · Study checks passed: the house unchanged in the design files (shift 0.91, 5.06 the outline asserted): study only · the calibration: the wall as built 0.72 m inside the deed line (B2's reading gives 0.68, the notch's 2.20 / 2.30 give 0.72 / 0.62: consistent within 0.10) · the current position measured as on site: the notch 2.20 m from the wall as built (= the 2.20 reading), B1 → the yard wall 4.77 (= the field value) · every option: a pure shift south (x unchanged, the sides and the diagonals unchanged), the driveway's 3.95 m untouched (the cornice 0.49 m clear), every corner inside the envelope · the recommended option (0.50 m) reaches ≥ 0.40 m from the wall as built at the lower reading (0.44), lies within the ground the owner checked (≤ 0.50) and has no conflict · no option conflicts with the palms, the garden's low wall, the guest path, the car bay, the gas cabinet, the tanks, the sewer's falls or the soakaways. Field values: the owner on site, day 1 (docs/SITE_SETOUT.md § 7 rows 18, 23, 24). Sources: base.typ (the site v2), design_gf.typ (the house), site_plan.typ (the services and the falls), roof.typ, bedroom1_bath.typ (B4), rotation_b2.typ and placement_steps.typ (the calibration method).